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#include <LPC21XX.H>
#include <string.h>

#define VERIFY_MODE    0
#define VOTE_MODE      1
#define CONFIRM_MODE   2
#define RESULT_MODE    3

#define SW1    (1<<1)    // DMK
#define SW2    (1<<3)    // ADMK
#define SW3    (1<<7)    // TVK
#define SW4    (1<<20)   // RESULT

#define LCD_D 0xff<<14
#define RS 1<<12
#define EN 1<<13

char Voter_ID[3][13] =
{
    "0600680EE989",
    "05005351C6C1",
    "060067F1C656"
};

void delay_ms(unsigned int ms);
void LCD_INIT(void);
void LCD_COMMAND(unsigned char);
void LCD_DATA(unsigned char);
void LCD_STRING(unsigned char *);

int voted[3] = {0,0,0};
int DMK=0, ADMK=0, TVK=0;

char system_state = VERIFY_MODE;
char selected_party = 0;
char verified_id[13];
char user[13];
int current_index = -1;

/* ===== UART CONFIG ===== */

void UART1_CONFIG(void)
{
    PINSEL0 |= 0x50000;    // P0.8 TXD1, P0.9 RXD1
    U1LCR = 0x83;
    U1DLL = 97;           // 9600 baud
    U1DLM = 0;
    U1LCR = 0x03;
}

int UART1_RX(void)
{
    if(U1LSR & 1)
        return U1RBR;
    else
        return -1;
}

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/* ===== RFID READ ===== */

int read_rfid(char *id)
{
    int data;
    int i=0;

    while(i<12)
    {
        data = UART1_RX();
        if(data==-1) continue;
        if(data=='\r' || data=='\n') continue;
        id[i++] = data;
    }
    id[i]='\0';
    return 1;
}

/* ===== CHECK VOTER ===== */

int check_voter(char *id)
{
    int i;
    for(i=0;i<3;i++)
    {
        if(strcmp(id,Voter_ID[i])==0)
        {
            current_index = i;
            if(voted[i]==1)
                return 2; // already voted
            else
                return 1; // valid
        }
    }
    return 0; // invalid
}

void LCD_STRING(unsigned char *s)
{
    while(*s)
        LCD_DATA(*s++);
}

void LCD_COMMAND(unsigned char cmd)
{
    IOCLR0 = LCD_D;
    IOSET0 = (cmd & 0xf0)<<10;
    IOCLR0 = RS;
    IOSET0 = EN;
    delay_ms(2);
    IOCLR0 = EN;

    IOCLR0 = LCD_D;
    IOSET0 = (cmd & 0x0f)<<14;
    IOCLR0 = RS;
    IOSET0 = EN;
    delay_ms(2);
    IOCLR0 = EN;
}

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void LCD_DATA(unsigned char data)
{
    IOCLR0 = LCD_D;
    IOSET0 = (data&0xf0)<<10;
    IOSET0 = RS;
    IOSET0 = EN;
    delay_ms(2);
    IOCLR0 = EN;

    IOCLR0 = LCD_D;
    IOSET0 = (data&0x0f)<<14;
    IOSET0 = RS;
    IOSET0 = EN;
    delay_ms(2);
    IOCLR0 = EN;
}

void LCD_INIT(void)
{
    IODIR0 |= LCD_D | RS | EN;
    delay_ms(20);

    LCD_COMMAND(0x01);
    LCD_COMMAND(0x01);
    LCD_COMMAND(0x0C);
    LCD_COMMAND(0x28);
    LCD_COMMAND(0x80);
}

void delay_ms(unsigned int ms)
{
    T0PR=15000-1;
    T0TCR=0x01;
    while(T0TC<ms);
    T0TCR=0x03;
    T0TCR=0x00;
}

/* ===== MAIN ===== */

int main()
{
    int check;

    LCD_INIT();
    UART1_CONFIG();

    IODIR0 &= ~(SW1|SW2|SW3|SW4); // switches input

    LCD_COMMAND(0x01);
    LCD_STRING("RFID VOTING");
    delay_ms(2000);

    while(1)
    {

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/* ===== VERIFY ===== */
if(system_state==VERIFY_MODE)
{
    LCD_COMMAND(0x01);
    LCD_STRING("SCAN CARD");

    read_rfid(user);

    check = check_voter(user);

    if(check==1)
    {
        LCD_COMMAND(0x01);
        LCD_STRING("VERIFIED");
        delay_ms(1000);
        strcpy(verified_id,user);
        system_state = VOTE_MODE;
    }
    else if(check==2)
    {
        LCD_COMMAND(0x01);
        LCD_STRING("ALREADY VOTED");
        delay_ms(1500);
    }
    else
    {
        LCD_COMMAND(0x01);
        LCD_STRING("INVALID ID");
        delay_ms(1500);
    }
}

/* ===== VOTE ===== */
if(system_state==VOTE_MODE)
{
    LCD_COMMAND(0x01);
    LCD_STRING("1.DMK 2.ADMK");
    LCD_COMMAND(0xC0);
    LCD_STRING("3.TVK 4.RESULT");

    while(system_state==VOTE_MODE)
    {
        if(!(IOPIN0 & SW1))
        {
            selected_party=1;
            system_state=CONFIRM_MODE;
        }
        else if(!(IOPIN0 & SW2))
        {
            selected_party=2;
            system_state=CONFIRM_MODE;
        }
        else if(!(IOPIN0 & SW3))
        {
            selected_party=3;
            system_state=CONFIRM_MODE;
        }
        else if(!(IOPIN0 & SW4))

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    {
        system_state=RESULT_MODE;
    }
}

/* ===== CONFIRM ===== */
if(system_state==CONFIRM_MODE)
{
    LCD_COMMAND(0x01);
    LCD_STRING("SCAN AGAIN");

    read_rfid(user);

    if(strcmp(user,verified_id)==0)
    {
        if(selected_party==1) DMK++;
        if(selected_party==2) ADMK++;
        if(selected_party==3) TVK++;

        voted[current_index]=1;

        LCD_COMMAND(0x01);
        LCD_STRING("VOTE SUCCESS");
    }
    else
    {
        LCD_COMMAND(0x01);
        LCD_STRING("MISMATCH");
    }

    delay_ms(1500);
    selected_party=0;
    system_state=VERIFY_MODE;
}

/* ===== RESULT ===== */
if(system_state==RESULT_MODE)
{
    LCD_COMMAND(0x01);
    LCD_STRING("RESULT");

    LCD_COMMAND(0xC0);

    if ( (DMK==ADMK) && (DMK==TVK) )
        LCD_STRING("DRAW");
    else if (DMK>ADMK && DMK>TVK)
        LCD_STRING("DMK WINS");
    else if (ADMK>DMK && ADMK>TVK)
        LCD_STRING("ADMK WINS");
    else
        LCD_STRING("TVK WINS");

    while(1);
}
}
}

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